Self-Service on a BigData Platform

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Intro

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Data Pipeline - ETL

Extract

Source and find the relevant data

Transform

Process and combine one or more data sources and get insights

Load

Store and present the insights

In Real-Time

Supporting multiple data formats

Able to handle millions of records / sec

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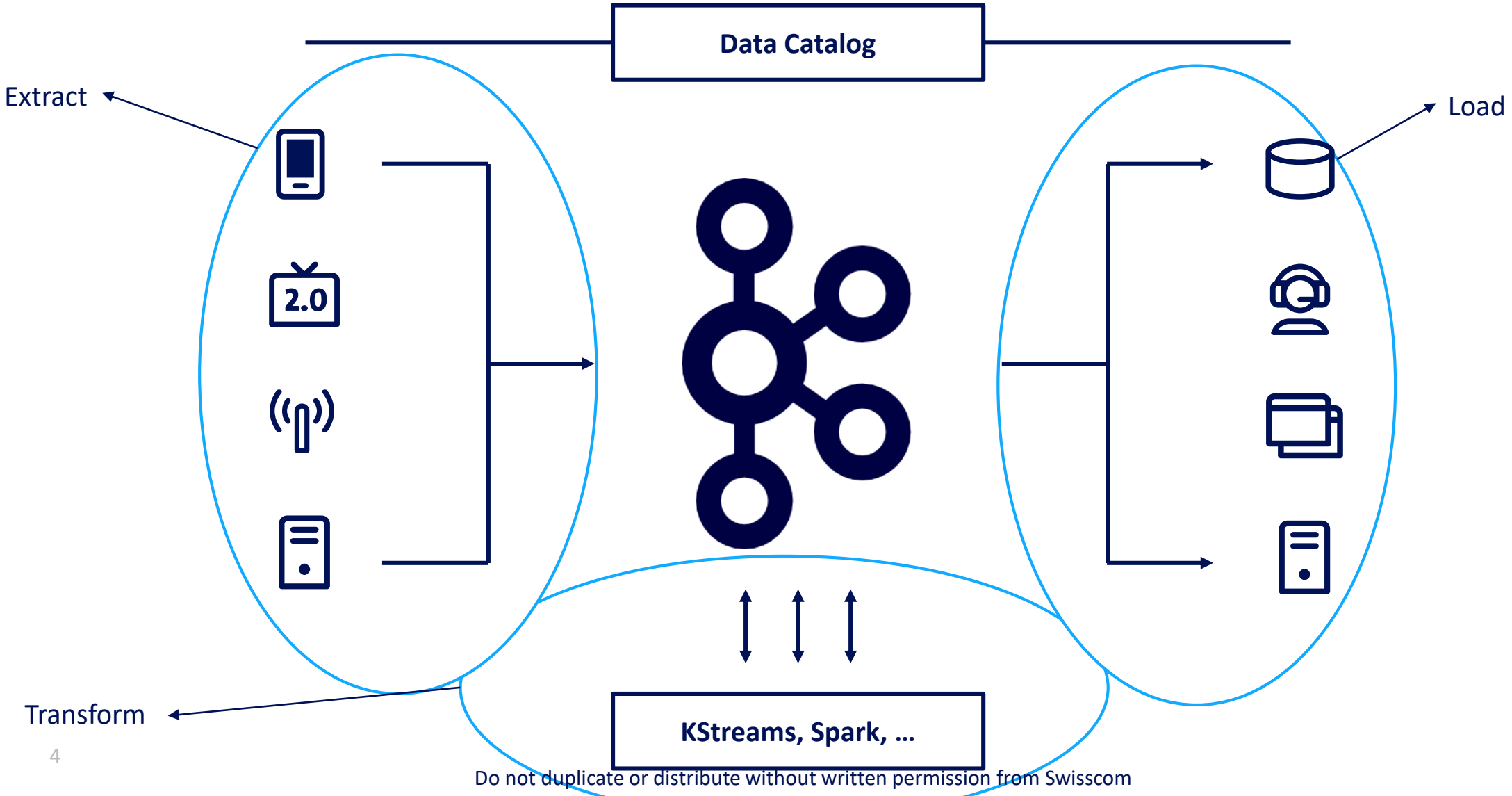
Swisscom's Streaming Platform



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History of BigData @ Swisscom

2014

Stream data into Hadoop

< 5 topics

< 5 users

2015

More data into Hadoop

10+ topics

< 5 users

200k messages/sec

2016

Stream Processing is a thing

100+ topics

30+ users

<1M messages/sec

2017

Role Based Access Control

1000+ topics

100+ users

>1M messages/sec

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How to scale

Challenges	Solutions
Lots of requests for topics/hdfs-path creation, modification, access, …	Self-Service API
Who does approve the access to a topic/hdfs-path?	Decentralized Approval Process
Policies are hard to maintain over time using existing tools	Define a higher-level Abstraction for Governance

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Governance

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Simple yet powerful concepts

Organization

Group of **individuals** working on something together (e.g. project), characterized by a **name**.

Space

Group of **isolated resources** (e.g. Topics) belonging to an organization, characterized by a **name** and a **classification**.

Resource

A resource is characterized by a **name**, a **description**, a **schema**, a **classification** (C1-C4), and **shareable flag**.

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Clear roles and responsibilities

Org. Owner

Responsible of the resources (usage, legal, approvals, etc.) **and the members** within his organization.

Org. Member

Use the resources within his organization.

Admin

Enforce the security and governance rules by technical means.

Auditor

Audit all the organizations to ensure rules are enforced and respected.

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How do we share resources between organizations/spaces?

Publish Resource

Owner of the resource publishes it in the data catalog

Request Access

Owner of an organization requests access to the resource

Approvals

Access to the resource is approved by its owner

Access

Resource virtually appears in the space and is readable by all identities

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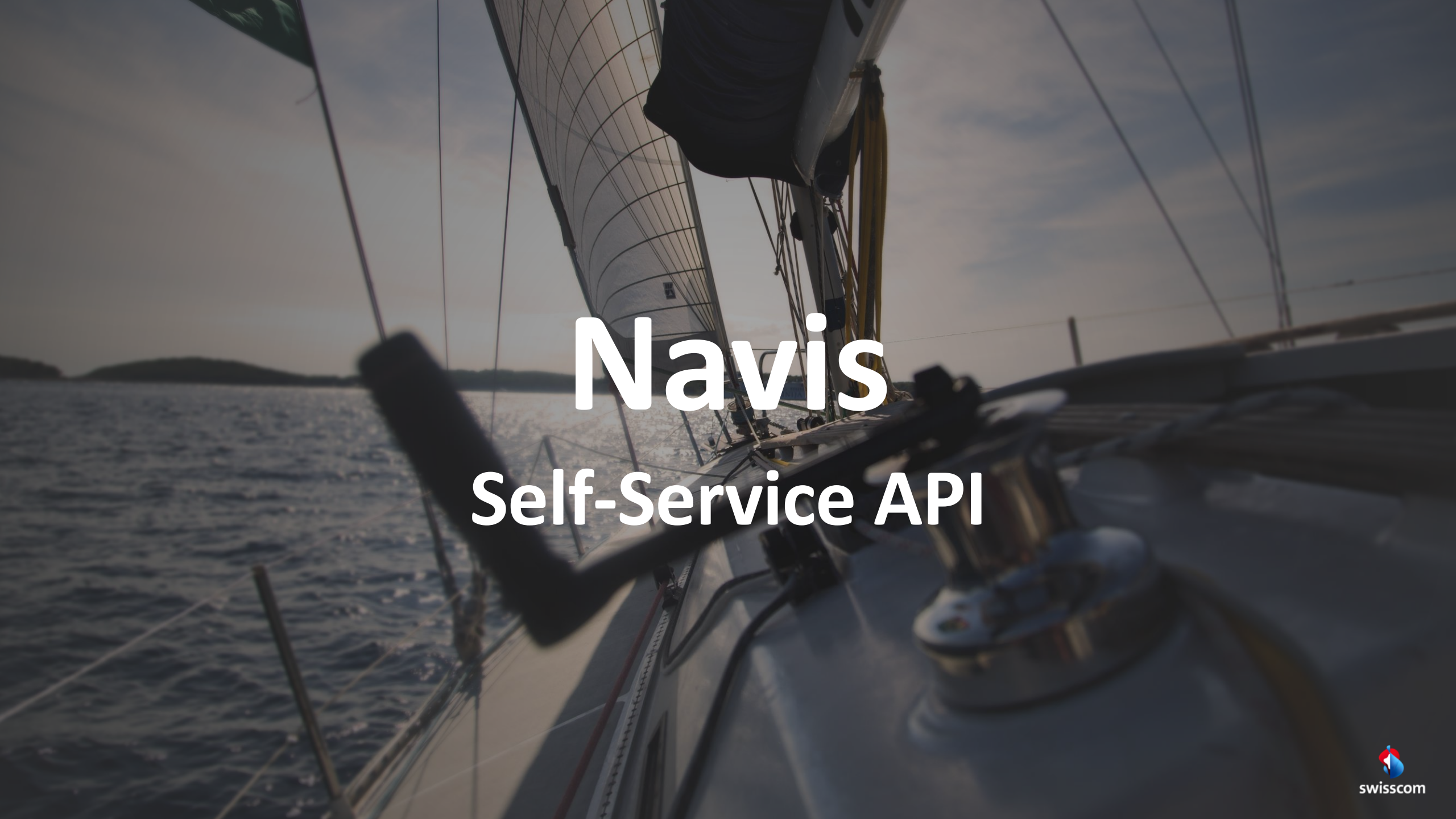

Navis

Self-Service API

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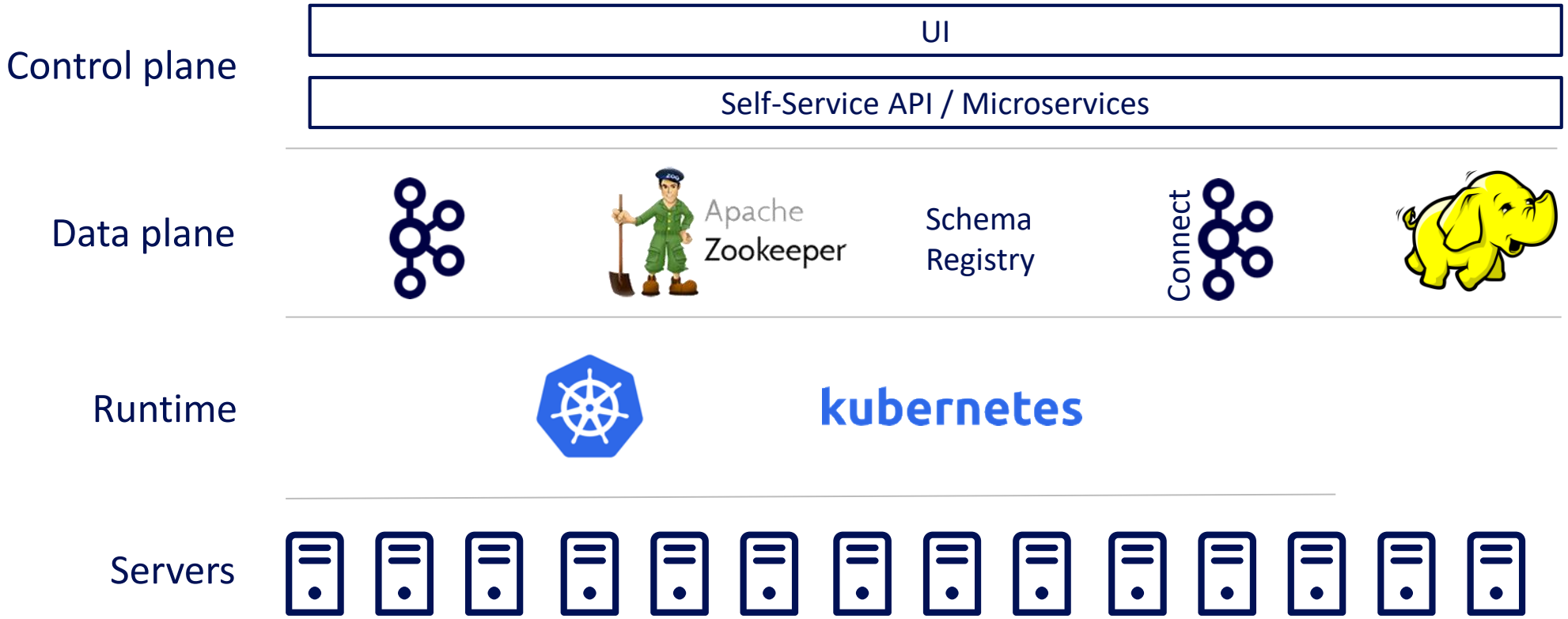
Architecture



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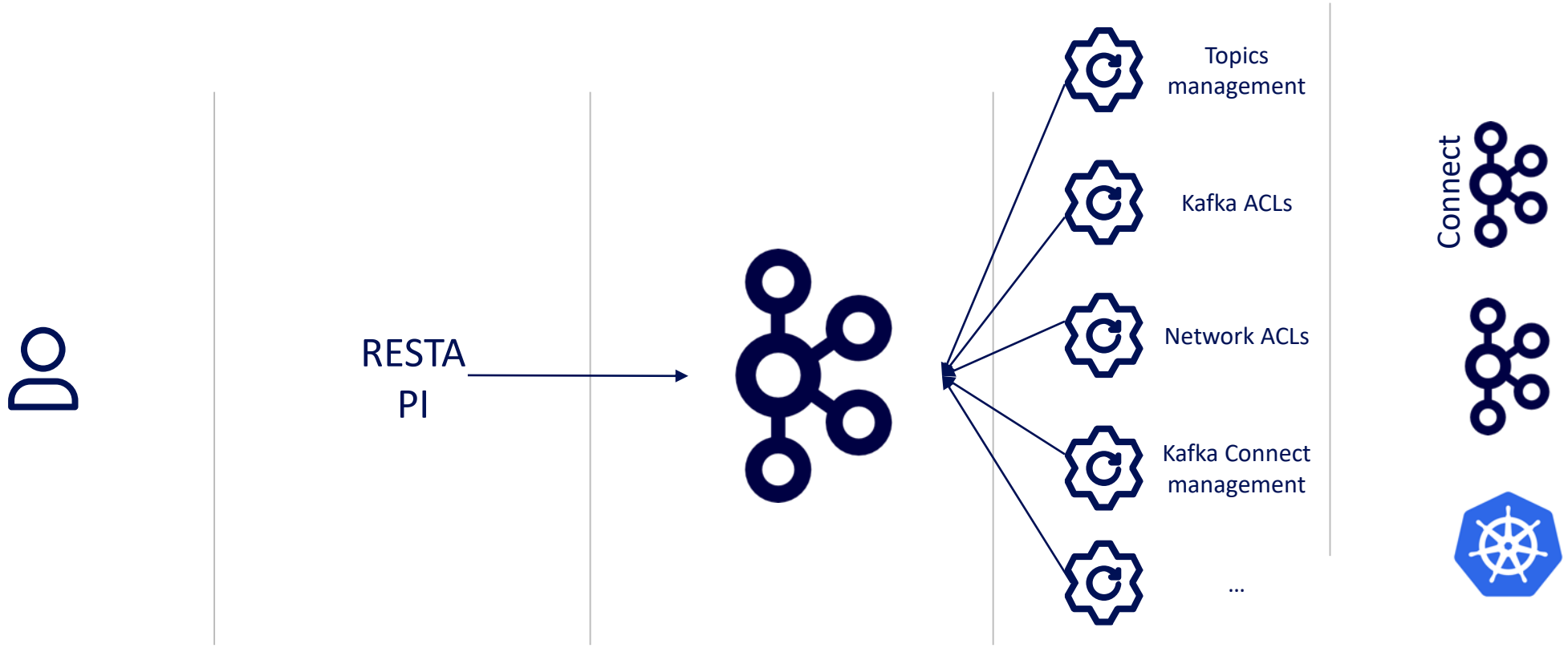
API Design



POST /organisations

API writes state to Kafka topics

State is stored in compacted topics

Microservices consume state from Kafka topics

Microservices talk to target systems to apply the state

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History of BigData @ Swisscom

2018

Kafka as a Service

1800+ topics
Topic as a Service
Connector as a Service

Hadoop as a Service

HDFS
YARN
Hive
Knox

2019

Streaming Platform as a Service

A lot of topics
A lot of connectors
Transformation as a Service
Multi-DC

Kubernetes as a Service

Secure Container Platform based on Kubernetes

?

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Kubernetes

as Enabler for a BigData Platform

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Kelsey Hightower
@kelseyhightower

Suivre

Kubernetes is a platform for building platforms. It's a better place to start; not the endgame.

13:04 - 27 nov. 2017

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What is Kubernetes?

Production-Grade Container Orchestration

Service discovery and load balancing

Automated rollouts and rollbacks

Self-healing

Horizontal scaling

Secret and Configuration Management

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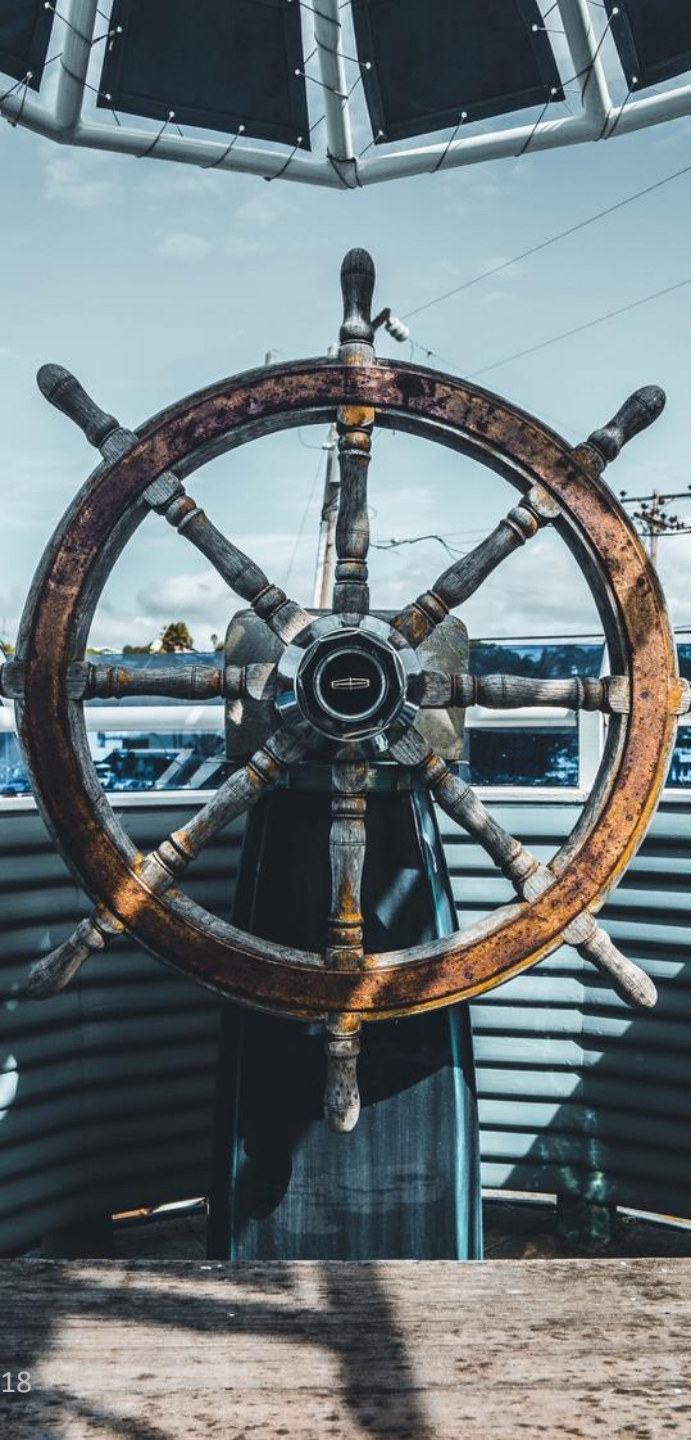

Why Kubernetes?

Scheduler

Network Policies

Containers

Increase hardware utilization

It rocks!

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Firehose 2.0

Data Streaming Platform

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Firehose Self-Service Entities

Kafka Topic

Dedicated Kafka Topics for every organisations for **ingesting data**.

Identity

Identity delivered to a Kafka Client, characterized by a **name**, a **network zone**, a **set of IPs**, and a **certificate**. Identity has RW access to all resources within its space.

Connector

A Connector will either **source** data from an external system to Kafka or **sink** data from Kafka to an external system.

Transformation

A Transformation is a **generic process** reading data from an input topic, processing the data and finally storing the result to an output topic.

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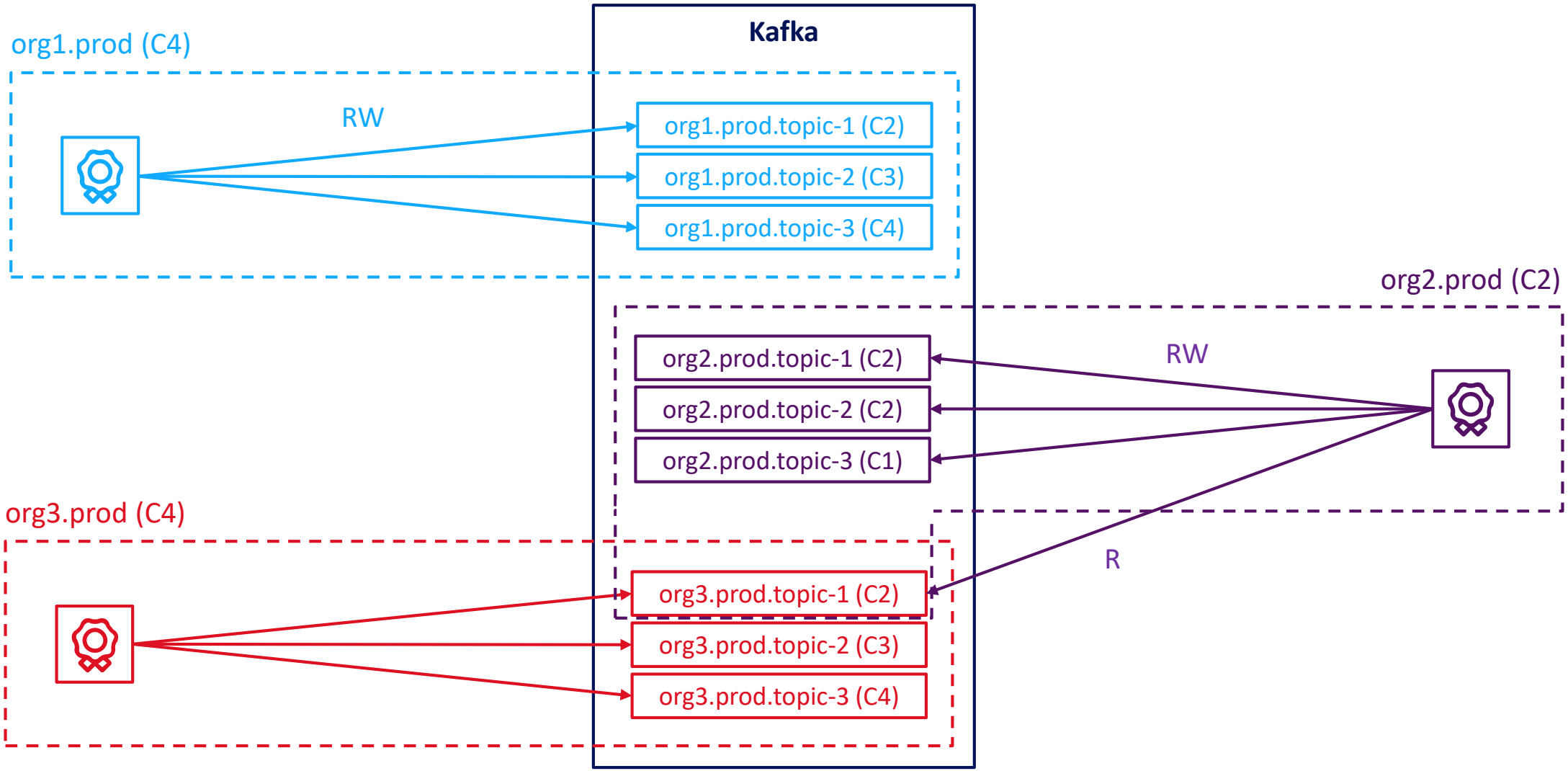

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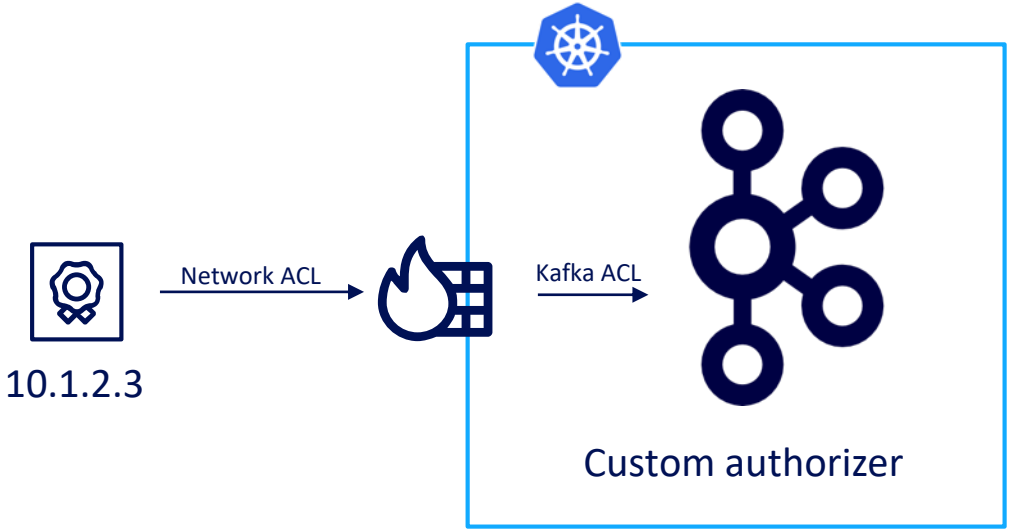
Kafka and Network ACLs

1. Identity created
2. Identity approved
3. Network ACL microservice:
   - creates Kubernetes network policy
4. Kafka ACL microservice:
   - adds Read/Write ACL for <org>.<space>.*
   - adds Read Only ACL for shared topics
5. User generates a certificate and starts to consume/produce to topics belonging to the space



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Connectors as a Service

Auto-Dump to HDFS

Automatically dump topics into HDFS

Users simply need to activate the feature and select some bucketing options

A connect cluster per topic is spawned on Kubernetes

Connectors as a Service

Connector images can be scheduled on Kubernetes via the REST API

Users need to select or create the connector image and configure the connector

Users are responsible to make sure the connector is running

We provide access to the connectors logs and alert users if a connector is failing

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Transformations as a Service

Supported Transformations

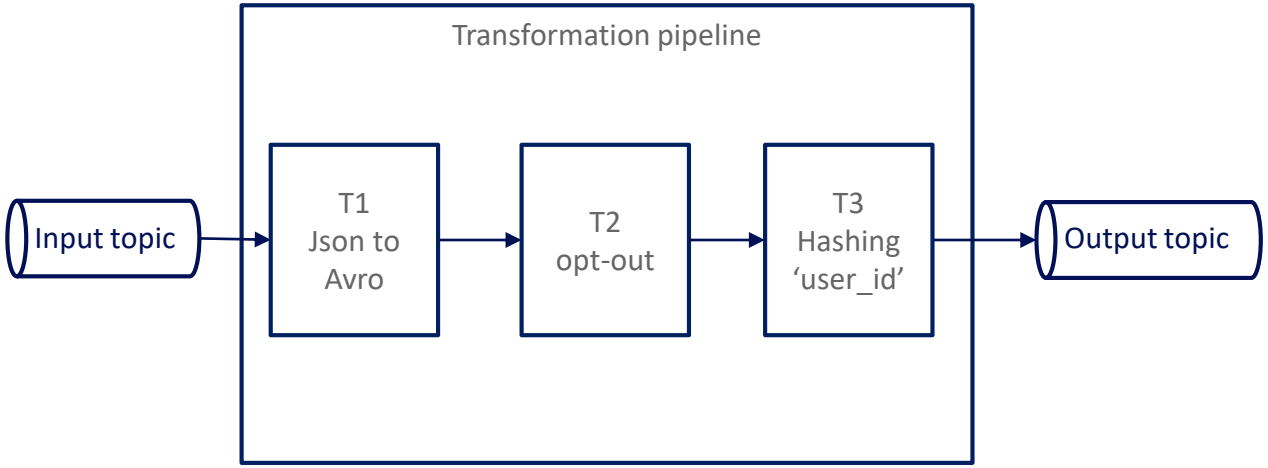
Changing format
e.g JSON to Avro

Hashing fields

Records filter (opt-in/out)

Fields filter

Transformation pipeline



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Hadoop

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Provided Hadoop Services

HDFS

The Hadoop HDFS service provides a **fault-tolerant**, distributed file system able to store data **reliably** even in the presence of failures including NameNode failures, DataNode failures and network partitions.

YARN

YARN is the **resource management** and **job scheduling** technology responsible for **allocating system resources** to the various **applications** running in a Hadoop cluster

Hive

Hive gives an **SQL-like** interface to query data stored in the Hadoop distributed file system

Edge Node

The edge nodes are **infrastructure components** accessible through SSH that **provide access** to Hadoop the services HDFS, YARN, Hive.

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Hadoop Self-Service Entities

HDFS Path

Dedicated **path** for every organisation for **storing data**

/data/*org*/*space*/resource_1
/data/*org*/*space*/resource_2

YARN Queues

Dedicated **processing power** and **memory** for organisations

org.application_1 (5%)
org.application_2 (10%)

Hive Tables

Dedicated Hive **databases** for the organization's **tables**

*org_space*_database.table_1
*org_space*_database.table_2

Edge Node

Dedicated **space** for storing, installing and **service users** running the organization's applications

/srv/home/srv-*org-space*

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What’s next?

Improved management of physical volumes and better integration with Kafka

Quotas for Kafka clients

Autoscaling

Data Lineage

Firehose in second data centre for geo-redundancy

Function as a Service

Container as a Service

Include other platforms in our new governance framework

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Thank you!

Bern, 28th of June 2019

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